# SSEMU SECONDARY SCHOOL

S.5 EOT 1 PAPER 2

Duration: 2hrs 45mins

**INSTRUCTIONS:**

**DON'T DARE FAIL THE MECHANICS QUESTION YOU HAVE CHOSEN BY YOUR OWN.**

Every response should start from a fresh page

## SECTION A (Attempt a number you can maximize)

**ITEM 1**

During a school mathematics project at Ssemu Secondary School, a group of S.3 students is tasked with designing a decorative but stable hexagonal metal frame to be mounted at the school's main gate. The structure is intended to support a hanging sign bearing the school motto.

The frame is shaped as a regular hexagon ABCDEF, fixed in a vertical plane. At each of the six vertices, tension forces act along wires connected to surrounding supports such as poles and nearby walls. Due to uneven fixing conditions, the magnitudes and directions of the forces differ slightly, creating a complex system that must be analyzed to ensure stability.

Mr. Colletta challenges the students to apply the knowledge of resolution of forces in a polygon to determine whether the structure is in equilibrium and, if not, how it can be adjusted, and also determine the line of action of the resultant.

At a particular instant, the following forces act at the vertices:

12 N along AB
10 N along BC
8 N along CD
6 N along DE
5 N along EF
9 N along FA

**TASK:** As a learner of applied mathematics, provide a well elaborated response to Mr. Colletta's challenge.

**ITEM 2**

At Ssemu Secondary School, a vertical radio mast is installed near the main gate. To keep it stable against strong winds, it is supported by two light inextensible cables attached from the top of the mast to points on the ground.

The cables exert forces on the top of the mast as follows:

Cable AB exerts a force of 80 N in the direction of bearing 045°.

Cable AC exerts a force of 60 N in the direction of bearing 135°.

A steady wind blows from the south, exerting a force of 50 N on the mast in the direction of bearing 000°.

The top of the mast is considered as a particle in equilibrium under the action of these forces.

**TASK:** As a learner of applied mathematics:

a) Determine the resultant force acting on the mast in component form.

b) Show that the forces are in equilibrium or otherwise determine the magnitude and direction of the unbalanced force.

## SECTION B (COMPULSORY)

**ITEM 3**

At Gen. Prof. Colletta Shoppers, the manager noticed that the number of students buying snacks during break time varies from day to day. To improve planning and reduce food wastage, she decided to record the number of samosas sold each day over a period of two weeks.

The data is shown below:

| No. of samosas sold (x) | No. of days (f) |
|---|---|
| 10 | 2 |
| 12 | 3 |
| 15 | 4 |
| 18 | 3 |
| 20 | 2 |

Task;

As a learner of applied Mathematics, help the manager inteprete the data by presenting ~~FINAL~~ to him the well calculated measures of ~~dispersion~~ central tendency.

**REMARK:**

DEMISTIFY THE MATHEMATICS AS MR. COLLETTA DID.

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